

Current research splits

This dated addendum supersedes every research-split date, year-specific evaluation reference and starting-notebook reference in the original roadmap that follows. The original proposal and its page numbers are retained as historical context.

Role	Inclusive NY session dates	Sessions / share
Development	2021-09-27 to 2024-09-30	786 / 60.23%
Validation	2024-10-01 to 2025-09-30	261 / 20.00%
Final Test	2025-10-01 to 2026-09-25	258 / 19.77%

Why these boundaries

The nearly five-year sample supports roughly three years for Development, one for Validation and the latest year for Final Test. Whole sessions and month-end boundaries preserve chronology and align with Silver storage. All 1,305 source sessions, including 14 empty sources, remain assigned; all 786,491,001 records are retained. Cutoffs were chosen from calendar duration, not outcomes.

Start here

notebooks/02_mbp1_development.ipynb

Use the GC MBP-1 (.venv) kernel and **scan_development**. Both holdouts are blocked by default. The historical ingestion notebook is locked by default. The proposed 02_exploration notebook has not been implemented; the new Development notebook is the tested starting workspace.

Evaluation rules

Tune with chronological folds within Development and fit transformations only on training folds. Freeze the candidate before Validation and the complete procedure before an explicitly authorized Final Test. Re-specify the roadmap's old year-based fold calendars before analysis. Purging and embargo must follow the declared horizons; no feature/label horizon is frozen by this addendum.

Authoritative records and scope

project_docs/mbp1_research_split_policy.md

configs/mbp1_research_splits.json

Policy: gc-mbp1-chronological-v1. The README, governance, data dictionary, historical report notices, access helpers and notebooks now reference this policy.

Verified: 28 unit tests and Ruff passed; the Development notebook ran successfully. No holdout event rows were read for this update. Original ingestion provenance and migration compatibility remain unchanged. Prior full-period integrity diagnostics remain disclosed. No feature, model, backtest or trading edge has been established.

Original roadmap: 30 September 2026. Its proposed research thresholds and mechanisms remain proposals, not an executed research contract. Original pagination follows this unnumbered policy addendum.

GC under the microscope

From a verified MBP-1 store to falsifiable forecasts, realistic execution tests, and a defensible go/no-go decision.

Recommendation: start with **02_exploration.ipynb**, after freezing a versioned research contract. Its first job is to establish which observations and measurements are trustworthy within Development. Then test a small number of mechanisms: best-quote imbalance, recent trade/book pressure, and the probability of near-term price instability.

The objective is evidence, not a compulsory trading strategy. A calibrated forecast of price direction, movement risk, or liquidity may be useful. A robust finding that it does not survive costs or out-of-sample testing is also a successful research outcome.

The recommended sequence

Phase	Question we must answer
1. Define and observe	What is a valid, available quote? What does this particular GC sample actually contain?
2. Measure and predict	Does an observable state improve a strictly future forecast beyond simple baselines?
3. Test economic value	Does that improvement survive delay, bid/ask execution, fees and changing conditions?
4. Confirm and operate	Does the frozen system generalize to 2024, then to a single authorized Final Test, then to prospective observation?

Research territory

The initial scope is the selected **GC.v.0** contract during **07:00-12:00 New York time**. MBP-1 is well suited to studying the visible best bid/ask and delivered trades. It cannot establish full-depth liquidity, individual queue position, hidden volume, trader identity, or a complete metaorder history.

The guiding book is *Trades, Quotes and Prices: Financial Markets Under the Microscope* by Bouchaud, Bonart, Donier and Gould. We will translate its mechanisms into GC-specific tests; equity-market results are hypotheses to examine, not facts to import. The reading sequence is on page 14. [1]

Status of this document: a proposed plan, not an executed research contract or an empirical result. Repository documentation, source code, notebook structure and primary references were reviewed. No market-data scans, feature fitting or Final Test research were performed for this roadmap.

STARTING POSITION

What is ready, and what is not

The engineering foundation is unusually strong. Research semantics and inference are the next work.

Verified project evidence	Implication for the roadmap
786,491,001 raw and Silver records; zero exclusions; 61 monthly partitions / 184 Parquet files.	Reuse the published Silver store. Do not repeat ingestion or rebuild a second canonical dataset.
Coverage: 2021-09-27 to 2026-09-25, selected weekdays and a five-hour daily window.	Any conclusion applies to this selection. Overnight behavior and full-day execution are outside the observed sample.
14 empty sessions, 61 warning sessions, six known vendor-degraded dates in the existing full-period audit.	Integrity does not imply research eligibility. These are prior infrastructure findings, not freshly inspected holdout statistics.
26 observed instrument-ID segments; physical expiration symbols unresolved.	Reset by session and observed roll segment. Do not model time to expiry or splice returns across contracts.
Bounded Arrow scanner; receive-time selection; native integers; Final Test guard.	Build additive research tools on this access path. The scanner does not implement research exclusions.

The unresolved issues that matter most

Event meaning. Historical channel-flush records may carry incorrect non-trade action/price/size. The current vendor issue board still describes that problem and distinguishes it from action=T records. Exact add/cancel/modify counts must therefore remain out of the initial feature set. [3]

Quote timing. The local contract conservatively distinguishes completed and partial events. Current vendor guidance also discusses normalized MBP handling. Verify the historical encoding, the role of F_LAST, and trade/quote sequencing with deterministic examples before declaring a quote pre-trade, post-trade or executable. [L2-L3, 4]

Clock meaning. Receive time is the ordering/availability anchor for this dataset; it is not our future trading system's arrival time. Negative event/receive differences must not be repaired by shifting clocks. [L2-L3]

Explicit negative finding: the repository contains no Gold feature layer, predictive model or backtest. No GC forecasting ability or trading edge has been demonstrated. Migration success is not a research result. [L1, L4-L5]

PROJECT MAP

One question per stage

Keep notebooks as research narratives; move reusable calculations into tested modules.

Order / artifact	Tasks and tangible output	Gate to continue
0 / Research contract	Freeze samples, eligibility, clocks, questions, labels, experiment budget and evaluation rules. Write the decision register.	No unresolved meaning in a primary feature or target.
1 / 02_exploration	Inspect Development eligibility, clocks, event structure and BBO distributions. Produce the observation map.	A defensible sample and measurement spec; no holdout access.
2 / 03_stylized_facts	Measure persistence, clustering, response and depth dependence. Produce mechanism cards, including null results.	Each candidate has a plausible mechanism and a falsifiable future target.
3 / Research dataset	Implement causal features, labels and bounded storage. Produce a versioned dataset and lineage manifest.	Chunk invariance, causality, boundary and sentinel tests pass.
4 / 04_hypothesis_tests	Test three registered questions with simple baselines and Development walk-forward splits.	Paired out-of-sample improvement with dependence-aware uncertainty.
5 / 05_prediction_models	Compare a small, fixed model ladder and ablations. Nominate one system.	Stable improvement attributable to a mechanism, not a search artifact.
6 / 06_execution_feasibility	Replay delayed bid/ask execution, costs and stress scenarios. Produce an economic decision memo.	Conservative economics and risk budget pass, or retain only forecast utility.
7 / 07_validation_2024	Run the frozen candidate and report all registered outcomes.	Pass the locked gate without retuning on 2024.
8 / 08_final_test	One separately authorized evaluation on the fixed 2025+ window.	Pass, fail or inconclusive; preserve the complete result.
9 / 09_paper_monitoring	Prospective observation, latency/fill measurement and drift monitoring.	Operational evidence before any live-capital decision.

The existing 01_mbp1_ingestion_and_validation.ipynb remains an infrastructure notebook. Do not use its Validation-day example or full-period quality plots as the seed of an exploration notebook. Proposed notebook names are new artifacts; none is created by this roadmap.

STAGE 0 / BEFORE EXPLORATION

Freeze the research contract

Write `project_docs/mbp1_research_contract_v1.md` and a matching machine-readable research configuration.

Separate the three time partitions

Partition	Permitted use
Development 2021-09-27 to 2023-12-31	All diagnostics, hypothesis development, fitting, model selection and execution-policy design, with chronological internal holdouts.
Validation 2024-01-01 to 2024-12-31	One frozen candidate evaluation. Inspection for this purpose begins only after the Development decision.
Final Test 2025-01-01 onward	Blocked by default. Recommend a fixed historical endpoint of 2026-09-25 for this data version; access requires a separate locked procedure and authorization.

Proposed initial policy - record it before executing

Universe and boundaries. Use only the selected GC stream and observed morning window. Reset rolling state at every session, instrument/roll change, invalid-book interval and material coverage break. Require a full 30-second history after resets for the common first feature set.

Primary research eligibility. Exclude known vendor-degraded sessions and empty sessions. Quarantine undefined, absent, nonpositive, locked or crossed books; invalid quantities; and relevant bad-time/book flags. Start with the project's conservative `F_LAST` book-state filter, subject to the semantic gate on page 6. Preserve every source row and a reason code in the derived eligibility view.

Coverage and freshness. Propose a reset for receive gaps greater than 60 seconds, matching the existing warning threshold. Use a maximum one-second quote age at a decision or target mark. A gap, invalid state or boundary inside a label interval makes that label unavailable; it must not become a zero return. These are proposed engineering choices, not optimized thresholds.

Sensitivity and selection. Primary analysis uses eligible intervals; a predeclared stricter check removes entire sessions with book/coverage warnings. Activity outliers remain warning flags, not automatic exclusions. Do not choose exclusion rules from their effect on P&L.

Freeze the search. Register three primary questions (page 8), their metrics, lookbacks, horizons, model grids and gates. Every attempted variant receives an ID, including discarded and negative trials. An amendment creates a new version and a declared new experiment before reading its outcomes.

Gate 0: research scope and measurement rules are unambiguous and versioned. This roadmap proposes defaults; it does not silently freeze or execute them. No source replacement, API purchase or migration-compatibility refresh belongs in this stage.

02_exploration.ipynb

A Development-only atlas of the observations we can legitimately use.

- 1. Reproducibility header.** Load the frozen contract and dataset identity. Assert start/end are within Development; record code/configuration hashes, environment, seed and CPU settings. Import the existing scanner and request only needed columns. No direct Parquet bypass or `allow_final_test` switch in this notebook.
- 2. Eligibility funnel.** Count delivered rows, candidate book states, valid states, decision samples, warm-up losses and label availability. Report overlapping reasons separately and also a mutually exclusive funnel whose counts reconcile. Use only Development rows from existing quality metadata.
- 3. Event and clock microscope.** Choose reproducible examples without outcome selection: first eligible day in each Development quarter, plus predeclared anomaly cases. Inspect bounded delivered sequences, equal timestamps, flags, T/F/N records, sentinel values and pre/post-event quotes. Keep `event_idx_day` as the tie breaker.
- 4. Contract and session map.** Display Development coverage and observed ID boundaries. Show distributions separately by session and segment; use a fixed New York time grid. Record where physical contract identity remains unknown. Sparse samples cannot establish low market activity versus missing upstream information.
- 5. Visible liquidity atlas.** Plot spread in ticks, both displayed sizes and order counts, size imbalance, quote age, valid quote-change rate and trade rate. Show medians and tails, time-of-day profiles, event-weighted and time-weighted views. Do not interpret repeated non-trade messages as fresh liquidity submissions.
- 6. Time-scale diagnostics.** On the predeclared grids, report midpoint-return distributions, zero-return share, movement size and observation survival. Compare 100 ms, 1 s and 5 s measurement scales; do not select the most attractive horizon. A midpoint can move by half a tick.
- 7. Research limitations.** List which measurements are supported, which are conditional and which are unavailable. In particular, quantify unknown trade-side coverage before using signed flow. Preserve duplicates; do not infer that identical trades are duplicate executions.
- 8. Decision page.** Produce a Development observation map, an eligibility report, a semantics memo and a short issue register. Promote only measurements whose interpretation is defensible. Store generated tables under `ignored data/metadata/mbp1_research/` and plots under `ignored outputs/`.

Gate 1: the notebook can run from a clean kernel with bounded memory; the sample reconciles; all charts identify their clock, denominator, partition and weighting. It ends with a measurement decision, not a list of visually attractive trading signals.

Suggested opening work package: contract and synthetic examples first; then one fixed Development day; then the full Development interval using streaming summaries. Do not scale an ambiguous calculation to hundreds of millions of rows.

MEASUREMENT SPECIFICATION

Make causality explicit

Most false microstructure discoveries begin with a target or information set that is slightly misaligned.

The observation contract

Process records in delivered order, keyed by session_date_ny and event_idx_day. A decision at receive-clock time t may use only completed information delivered strictly before t . Equal-time records retain delivery order. Event time is a secondary diagnostic clock; it must never pull a later-received message into an earlier decision.

Build a derived book-state stream, a separate action= T trade stream, and a fixed 100 ms decision grid. An as-of state is the last eligible observed state, with an explicit age; it creates no synthetic source event. Invalidate it across breaks. BBO-state transitions are measurements of visible state, not reconstructed individual orders.

Required semantic check: reconcile the frozen local F_LAST convention with the historical decoder and current vendor documentation. Do not assume a trade record's attached BBO is the pre-trade quote. Test complete event groups, same-timestamp records and channel-flush examples before aligning trade response. If unresolved, proceed only with measurements that do not depend on the disputed sequencing. [L2-L3, 2-4]

Minimal, interpretable features

Quantity	Definition and safeguard
Midpoint / spread	Let b, a be valid native bid/ask prices; $T=100,000,000$ native units per GC tick. Store $M2=b+a$ exactly; $spread=(a-b)/T$. Compute returns as changes in $M2$ divided by $2T$. Check integer overflow before arithmetic.
Queue imbalance	$I=(q_bid-q_ask)/(q_bid+q_ask)$. Use defined positive sizes only. It describes visible best-level pressure, not hidden liquidity or our queue position.
Weighted midpoint proxy	$w=(a*q_bid+b*q_ask)/(q_bid+q_ask)$. Its displacement from midpoint in ticks is $I*spread/2$. Use this identity to avoid overflow; it is not independent of imbalance or a calibrated fair price.
Trade pressure	Past signed action= T quantity over 1 s and 5 s; B=buy, A=sell. N remains unknown. Keep unknown volume share and a predeclared minimum 90% known-side share for normalized signed-volume features.
Book pressure / risk	Changes in visible best-level price/size, past 1 s and 5 s pressure, 30 s realized movement, spread, depth, quote age and time of day. No exact cancellation or submission intensity.

Normalize or standardize only using each training fold. Retain exact native state for audit; floating-point statistics belong in the derived research layer. Trade-side and quantity conventions must match the source rather than a quote-based sign guess. [2, 4]

STAGE 2 / MECHANISMS

03_stylized_facts.ipynb

Use the book to organize measurements, then ask whether each relationship reaches into the future.

Price formation and tick regime. Measure the time spent at one-tick versus wider spreads, midpoint and trade-price return autocorrelation, and realized variation at the fixed sampling scales. Distinguish bid/ask bounce from midpoint dynamics. Decide whether a large-tick approximation is empirically reasonable for each Development stratum; do not label GC large-tick by analogy alone. Read chapters 2-7. [1]

Trade-flow persistence. For signed trades, estimate the correlation of signs at predeclared lags 1, 2, 5, 10, 20, 50 and 100, plus a simple next-sign baseline. Report unknown-side coverage. Trade-sign predictability can coexist with little future return predictability. This is a mechanism diagnostic, not proof of directional alpha. Read chapter 10. [1]

Event clustering. Compare inter-trade and valid quote-change durations, count dispersion, and past activity versus subsequent activity. First remove predictable time-of-day variation using training-only estimates. Fit an interpretable intensity baseline before considering a Hawkes process. Read chapter 9. [1]

Trade response and liquidity. Estimate average signed midpoint response from a verified pre-trade state to 100 ms, 1 s and 5 s later, stratified by pre-trade depth/spread. Then separately forecast from the first usable post-trade decision state. The first measures association around a trade; only the second tests available prediction. Read chapters 11 and 13-14. [1]

State-based order-flow imbalance. Construct net best-level pressure from adjacent eligible quote states. At unchanged prices this reduces to the bid-size change minus the ask-size change; price changes require the standard price-comparison terms. Aggregate only past transitions. Refer to the Cont-Kukanov-Stoikov construction, but label our completed-state measure as an adaptation, not a recovery of all underlying events. [7]

Liquidity adjustment and adverse selection. Test whether persistent buying is accompanied by changing depth, spread or weaker incremental response. Examine future signed markouts from observable states. They inform the cost of providing liquidity but do not measure a strategy's realized passive-fill P&L. Read chapters 16-17. [1]

Outputs and exit decision

Create one mechanism card per study: economic story, precise information set, estimator, sample count in sessions as well as rows, expected direction, uncertainty, alternative explanation and explicit negative result. Report a predeclared set of time-of-day and segment summaries; do not promote the best subgroup after inspection.

Gate 2: promote a mechanism only if it yields a testable forward question and can be measured reliably. A strong contemporaneous pressure/price relation is not enough. If the baseline effects are unstable or data-dependent, document that and reduce scope before adding model complexity.

STAGE 4 DESIGN / REGISTER BEFORE DATA

Three initial prediction questions

All numbers below are proposed starting specifications. Freeze them in Stage 0 before their outcomes are inspected.

Question	Target and baseline	Evidence sought
Q1 / Best-quote pressure	At the fixed 100 ms grid, predict the first nonzero midpoint move within 5 s: up, down, or no move by 5 s. Baseline: training-only class probabilities conditioned on the fixed time bucket and spread.	Does size imbalance add calibrated information? Start with a simple multinomial logistic model; use paired log-loss gain as primary metric. [6]
Q2 / Future price change	Predict the 1 s midpoint change in ticks using past 1 s / 5 s signed trades and state pressure. Baselines: zero change, then a controls-only linear model.	Does flow add forward information beyond imbalance, spread, depth, past movement and time of day? Primary metric: paired MSE improvement.
Q3 / Movement risk	Predict whether absolute 5 s endpoint midpoint displacement is at least 2 ticks. Baseline: time-of-day plus lagged 30 s movement, spread and depth.	Do liquidity and activity features improve a calibrated risk probability? Primary metric: paired Brier-score improvement. This target is endpoint movement, not maximum excursion.

Label construction rules

At t , use the last valid complete book strictly before t . At $t+h$, use the last valid book available by that receive-clock cutoff, with age at most 1 s. Never borrow the first later quote to fill a stale target. Reject a label if its observation interval crosses a reset, gap, invalid book, session end or roll. Save decision and target event indices.

For Q1, inspect only subsequent completed eligible states. A fully observed 5 s interval with no change is the no-move class; a truncated or invalid interval is censored and reported separately. Do not discard no-move cases to inflate hit rate. Restrict decisions so the full horizon lies before 12:00.

Limit the search before it starts

Primary horizons are the ones above. Predeclare 100 ms and 5 s return diagnostics for Q2 as secondary sensitivity checks; report them all. Use one feature definition per mechanism before testing alternatives. Do not scan hundreds of thresholds, indicators, horizons or hand-picked news dates.

Promotion rule: compare a candidate against its relevant baseline on identical eligible rows and sessions. A feature family advances only with a positive, practically relevant out-of-sample gain and acceptable stability. Three primary questions form one confirmatory family; secondary findings remain exploratory until independently confirmed.

Possible usefulness is broader than trading direction: Q3 may support abstention or position-risk limits. It still requires evaluation against a concrete decision baseline before being called economically valuable.

STAGE 3 / QUANT DEVELOPMENT

Build a small, causal research layer

The Gold layer is an additive, versioned view. The Bronze and Silver foundation remains authoritative.

Proposed module responsibilities

Location	Responsibility
src/research/eligibility.py	Split checks, book validity, quality reasons, clock rules and reset boundaries.
src/research/events.py	Completed-state and trade views; canonical ordering; carry state across Arrow batches.
src/research/features.py	Past-only features, exact price arithmetic and explicit lookback availability.
src/research/labels.py	Bounded future-label queue, censoring, as-of endpoints and no boundary crossing.
src/research/evaluation.py	Walk-forward splits, paired baseline scoring, clustered uncertainty and experiment ledger.
src/research/replay.py	Position/order state, simulated delays, bid/ask fills, costs and risk accounting.

Dataset design and reproducibility

Keep one row per eligible 100 ms decision, plus small event-indexed diagnostic views. Five hours give at most 180,000 grid points per complete session before exclusions. Stream event history rather than loading the full store into pandas.

Carry rolling state across batch and file boundaries within the same valid segment; reset only at specified market/data boundaries. Write date-partitioned research Parquet under ignored data/processed/mbp1_research/<version>/. Store contract hash, Silver identity, code revision, feature/label versions and per-reason counts in an ignored research manifest. Keep test labels physically separate from predictor inputs.

Use the existing environment, deterministic CPU numerics, fixed seeds and thread settings. Benchmark a bounded Development sample and record peak memory before scaling.

Tests that protect the research claim

Use hand-computable synthetic streams for same-time records, partial groups, unknown sides, identical trades, sentinels, crossed quotes, gaps, warm-up and rolls. Verify identical outputs across chunk sizes 1, 7 and 250,000. A future-record perturbation must not change earlier features; a prefix run must reproduce all features for the same prefix. Verify label signs and half-tick changes manually.

Gate 3: reconciliation, causality, chunk invariance and boundary tests pass. Before editing existing symbols run GitNexus impact analysis and report the blast radius. Run unittest discovery and Ruff after changes; run detect_changes before any commit. Never refresh migration compatibility to make a mismatch disappear.

STAGES 4-5 / QUANT RESEARCH

From baselines to one candidate

04_hypothesis_tests.ipynb establishes evidence; 05_prediction_models.ipynb tests whether modest complexity helps.

Development evaluation schedule

Proposed expanding-window folds: train through June 2022, score July-December 2022; train through December 2022, score January-June 2023; train through June 2023, score July-December 2023. Reserve the last training month inside each fold for tuning/calibration, with earlier data for fitting. Purge any training label whose horizon intersects a held-out interval; keep sessions intact and reset histories at their starts.

Freeze this schedule before running it. Report each fold, session-weighted aggregate performance, and event-weighted results as a sensitivity. Avoid random row splits: nearby features and overlapping labels are highly dependent.

Model ladder and search budget

First rung. Training-only unconditional/seasonal baselines; Q1 multinomial logistic regression, Q2 ridge regression, Q3 logistic regression. Fit scaling, clipping limits and calibration only inside training data. No whole-period normalization.

Second rung. A fixed piecewise imbalance curve or a shallow boosted-tree model only if linear residuals suggest a specific interaction. Prefer the current environment's available tools; a new dependency is a separately reviewed change. Deep networks and automated broad searches are deferred.

Budget. Propose at most 12 prespecified configurations per primary question, including model/hyperparameter variants, and four named ablations: controls only; add imbalance; add trades; add state pressure. Log the actual total search history. Do not hide failed trials when reporting the winner.

Inference and practical significance

Compare losses pairwise on the same decisions. Aggregate to sessions and use a fixed-seed moving-block bootstrap of five consecutive sessions, with 2,000 resamples, for uncertainty; individual events are not independent trials. Check a prespecified 20-session block sensitivity. Use Holm adjustment at familywise 5% across the three primary questions. Development intervals remain exploratory after model search; multiplicity adjustment does not remove winner-selection bias. The frozen 2024 evaluation supplies independent confirmation.

Propose a screening floor of at least 1% relative reduction in the primary loss, a 95% interval for paired improvement above zero, and the same improvement sign in at least two of three folds. Treat a benefit that disappears under the longer-block sensitivity as inconclusive. These are research governance choices, not established GC effect sizes.

Gates 4-5: nominate at most one candidate system and retain its simplest credible comparator. Include calibration, ablations, sample coverage and null findings. If gains are tiny, unstable, or explained by a time-of-day baseline, stop that line of research rather than trying ever more models.

STAGE 6 / BEFORE EXTERNAL VALIDATION

06_execution_feasibility.ipynb

Determine whether a forecast is usable under plausible execution, rather than equating statistical accuracy with profit.

- 1. Freeze a minimal policy.** Start with one contract, at most one open position, no pyramiding, and a predeclared entry threshold, exit horizon and abstention rule derived only within Development. Close before the observed window ends. Session and roll boundaries are hard limits, not synthetic trades.
- 2. Replay information and delay.** A decision uses the state available at t ; order arrival is t plus a stated latency. Propose 10, 50 and 250 ms scenarios, with 50 ms as the research base case until a measured system latency replaces it. The zero-delay case is a diagnostic upper bound. Vendor `ts_recv` alone does not validate a live latency assumption.
- 3. Use bid/ask economics.** Simulated buys cross the ask and sells cross the bid at arrival, subject to valid quote age and displayed size. Mark exits on the executable side. A long round trip earns $100 \times (\text{exit bid} - \text{entry ask})$ dollars per GC contract before fees and additional slippage; a 0.10 price tick is \$10. Bid/ask fills already charge the spread: do not subtract it twice. [8]
- 4. Separate known costs from assumptions.** Use a dated, applicable broker/exchange fee schedule before making an economic claim. Until then, report break-even round-trip cost and a transparent stress grid. Propose 0, 0.5 and 1 extra tick per side for slippage beyond observed quotes; the fractional value is an expected-cost scenario, not a legal execution price.
- 5. Make missing execution visible.** No touch-equals-fill rule for passive orders. No fills larger than observed top-level quantity and no invented sweeps through unseen depth. If entry is unavailable, record an unfilled decision. If a held position loses valid exit data, record an unresolved execution and adverse bounds; never delete the trade from P&L. Material unresolved exits block an economic pass.
- 6. Measure robustness and concentration.** Report gross/net ticks, net dollars, turnover, time in position, maximum drawdown, losing streaks and P&L by session. Include latency/cost surfaces and leave-one-month-out contribution diagnostics. Report all trades and failures, including low-confidence decisions and abstentions.

Gate 6: for a trading candidate, require positive net expectation and a positive lower 95% session-block interval under the locked base cost/latency case, plus compliance with a numerical drawdown/loss budget fixed before 2024. Report stress failures explicitly. If costs erase the edge, retain the forecasting result without calling it a trading system.

MBP-1 can support a small-size aggressive-execution approximation. Exact passive queue fills, impact at scale and strategy-induced market response are not identifiable here. Any market-making result must remain a scenario study until richer order/fill evidence is available.

STAGE 7 / HELD-OUT YEAR

07_validation_2024.ipynb

A confirmation exercise with a frozen artifact and an explicit failure path.

Before opening 2024

Archive the candidate's research contract, features, fitting recipe, model parameters, calibration, decision thresholds, execution assumptions, eligibility policy, expected sample rules and pass criteria. Hash the executable artifacts. Refit using Development only under the already selected recipe; do not tune from 2024 diagnostics.

Recommended first protocol: one fixed model for all of 2024, with no label-driven online adaptation. A later adaptive model is a different experiment whose update schedule must be defined beforehand. This makes a failure easier to interpret and a success easier to trust.

Run exactly the registered report

Data and availability. Apply the same rules without manual exceptions, including the already documented degraded-date overlay. Report eligible coverage, censored targets, quote age and unfilled/unresolved executions. Unexpected quality failures cause a recorded pause, not a new profitable exclusion.

Prediction. Report the candidate versus its frozen baselines, calibration, paired primary loss, uncertainty and all prespecified horizon checks. Evaluate the same target definition used in Development. Keep favorable subgroups from becoming a replacement primary result.

Economic value. For the trading branch, run the fixed policy and costs, including drawdown, session concentration and delay stress. For a risk/abstention branch, score its predeclared decision utility against an equally constrained baseline; forecasting gain alone is insufficient to claim trading value.

Stability. Use predeclared calendar quarters, time-of-day buckets, spread states and observed contract segments. These are diagnostics; if one of them becomes a new selected strategy, 2024 has been used for development and is no longer an untouched validation set for that strategy.

The decision is recorded once

Outcome	Next action
Pass	Same-direction, practically meaningful primary gain (proposed 1% floor and interval above zero), required decision/economic gate passes, no measurement breach: prepare the locked Final Test package.
Inconclusive	Coverage or precision is inadequate, or stress results are fragile: do not consume Final Test to settle design questions. Preserve the result and seek prospective evidence.
Fail	No gain, broken calibration, negative economics or an invalid assumption: document failure. Any redesign is a new research version, with 2024 explicitly marked as consumed.

Gate 7: only a fully specified candidate that survives the predeclared 2024 decision advances. No repeated 'validation' runs that tune thresholds until the year becomes profitable. Final Test remains blocked throughout this stage.

STAGES 8-9 / CONFIRMATION AND OPERATION

A protected test, then prospective evidence

08_final_test.ipynb is a release procedure; 09_paper_monitoring.ipynb is an operational research record.

Stage 8: one authorized Final Test

Fix the window at 2025-01-01 through the current source endpoint, 2026-09-25, before access. Use the same frozen Development-trained model that passed 2024 for the first campaign; do not silently refit on the validation labels. Version any alternative refitting protocol as a new design decision before release.

Create an isolated evaluation entry point with an exact date allowlist, immutable configuration and no tuning controls. Only the separately authorized procedure may set `allow_final_test=True`. Direct file access remains governed too: a scanner flag is not a complete security boundary. Log every holdout access.

Evaluate the full registered window once, without interim outcome inspection, optional stopping or moving the endpoint. Produce the same coverage, prediction, calibration, economic and stability report as in 2024. Existing full-period infrastructure diagnostics do not justify choosing features from 2025+.

Gate 8: pass only against criteria frozen before access. If it fails, preserve the negative result and retire the claim. A revised model may not reuse the same period as an untouched Final Test; it needs genuinely new prospective evidence.

Stage 9: prospective shadow/paper operation

Connect research to an observable system. When separately provisioned and authorized, record feed availability, decision timestamps, compute latency, intended orders, acknowledgements and actual/paper fills. Map historical and live schema differences explicitly; do not copy a historical action workaround into live data by assumption.

Fix a pilot protocol before starting. Propose a minimum 60 eligible sessions with the frozen system, plus a prespecified precision requirement. Review on a fixed schedule; risk stops remain active. If precision is inadequate, call it inconclusive rather than watching until profit appears.

Monitor actionable failures. Check quote age, gaps, schema/order violations, model-input range, calibration drift, realized slippage and cost drift. Define kill switches and an investigation/restart checklist. Monitor labels only after they mature; never rewrite historical model decisions.

Decide whether richer data is justified. If passive execution or impact dominates the uncertainty, scope a separate MBO/depth/fill-data project. If the core forecast fails, more data purchase is not an automatic next step. No credentials, downloads or billable calls are part of this roadmap.

Live-capital deployment is a separate decision requiring resolved physical contract mapping, explicit risk limits and operational evidence. A favorable historical PDF is not sufficient evidence to deploy.

READING PLAN

Use the book as a research spine

Read a small chapter group, implement its measurement discipline, then record what does and does not transfer to GC.

Reading sequence [1]	Project use	Boundary of the claim
Chapters 1-4 Market setting, price statistics and the book	Stages 0-1: define observed prices, spread, depth, sampling clocks and the descriptive atlas.	Our five-hour futures sample is not a full-session Nasdaq equity sample.
Chapters 5-7 Single and joint queue behavior	Stages 2 and 4: motivate imbalance, competing price-move directions and duration questions.	Visible aggregate states do not reveal exact order lifetimes or cancellation hazards.
Chapters 9-10 Clustering and persistent flow	Stage 2: measure activity clustering and trade-sign dependence before fitting a point process.	Clock seasonality and correlated sampling can imitate more elaborate dynamics.
Chapters 11, 13-14 Trade response and dynamic impact	Stages 2 and 4: separate contemporaneous response from forward residual prediction.	Observed response is not a causal estimate of our own order's market impact.
Chapters 16-17 Spread economics and liquidity provision	Stage 6: interpret markouts, adverse selection and cost hurdles.	Queue position and passive execution are missing; profitability is not identified by markouts.
Chapters 8, 12, 15 and later parts Broader models and execution themes	Later study after the first evidence cycle; use theoretical models to explain residual questions.	Do not make full-book simulation, metaorder impact or optimal execution prerequisites for the first defensible forecast.

What to postpone deliberately

Defer add/cancel/modify decomposition until source semantics are resolved; passive market-making optimization until fills are defensible; Hawkes/propagator estimation until simple empirical diagnostics work; and deep learning until a compact baseline survives outside Development.

Also defer macro-event effects, cross-market lead/lag and contract-expiry conditioning. They require independently timestamped calendars or other instruments, aligned information clocks, resolved contracts and a new frozen question. Time-of-day activity near a scheduled release is not itself a verified news label.

Reading deliverable: a one-page note per chapter group: mechanism, measurable variables, assumptions, proposed falsification and GC result or null. The publisher's contents were checked for this roadmap; a full local copy of the book was not supplied, so this is a reading map rather than a claim to reproduce its detailed derivations.

IMMEDIATE WORK PLAN

The first research cycle

Complete these work packages in order; scale the workload only after the preceding gate passes.

Work package	Concrete tasks	Definition of done
A / Contract and semantics	Draft the research contract and three question cards. Fix eligibility, grid, horizons, thresholds and experiment budget. Review tiny synthetic event/quote examples.	A versioned specification, open-issue list and no ambiguous primary measurement.
B / Exploration skeleton	Create 02_exploration with a Development guard, configuration header, bounded loader, audit funnel and fixed example selection.	Clean-kernel run on one fixed Development day; no other partition accessed.
C / Measurement layer	Implement the state/trade views, clock grid, past-only feature history and censor-aware labels. Test chunk boundaries and hand-computed examples.	Tests pass, memory bounded, lineage and eligibility counts reconcile.
D / Development atlas	Expand the same notebook to Development; produce the predefined liquidity, timing and sampling diagnostics.	Observation map and explicit support/reject/defer decisions for each measurement.
E / First mechanism test	Complete 03_stylized_facts and its mechanism cards; then test Q1 imbalance against its baseline using the registered folds.	A reproducible positive, null or inconclusive finding with calibration and session-level uncertainty.

How the two roles divide the work

Quant developer: protect ordering and provenance; implement reusable causal calculations; enforce split guards; test boundaries; control memory; make each run reproducible. **Quant researcher:** define the mechanism and estimand; challenge sample selection; separate description from prediction; quantify uncertainty; decide whether an effect is useful after costs.

The decision register

For every experiment save: ID and version; question; economic mechanism; data and eligibility hashes; information cutoff; feature and target definitions; attempted variants; baseline; evaluation periods; metrics and intervals; economic assumptions; result; and the decision to promote, reject or defer. Record the reason for rejection in plain language.

The next concrete action: write the Stage 0 contract and scaffold 02_exploration.ipynb. Its first milestone is a trusted Development observation map. No model family, backtest engine or new data purchase should precede that milestone.

Schedule by gates rather than promising dates before runtime is measured. The first bounded run should establish the cost of a full Development pass; source-semantics questions may take longer than implementation. Freeze a small workload and finish its evidence cycle before expanding the project.

EVIDENCE AND REFERENCES

Sources and interpretation

Project facts come from the existing repository. External sources inform definitions and proposed tests, not claims of a GC edge.

Local evidence reviewed

[L1] README.md: scope, current foundation, access and operational constraints.

[L2] project_docs/mbp1_ingestion_contract.md: immutable representation, clocks, quality rules and governance.

[L3] project_docs/mbp1_data_dictionary.md: native schema, flags, fields, access and limitations.

[L4] project_docs/mbp1_foundation_findings.md: completed integrity/quality evidence and unresolved research semantics.

[L5] project_docs/mbp1_migration_report.md: independent environment and migration evidence.

[L6] AGENTS.md; src/data/mbp1_access.py; notebooks/01_mbp1_ingestion_and_validation.ipynb: required controls and actual entry points.

Repository snapshot reviewed: commit 939f054. GitNexus was used to inspect the access flow and scan_mbp1 callers.

Infrastructure counts are inherited audit findings, not new research scans. The 22-test migration result applies to this repository; the older 716-test figure belongs to the source project's larger suite. [L4-L5]

Primary external references

[1] **Bouchaud, Bonart, Donier & Gould (2018), Trades, Quotes and Prices** Cambridge University Press. Publisher contents verified; the chapter-to-project mapping is a proposed research design.

[Open primary source](#)

[2] **Databento: MBP-1 schema** Best-level coverage, native prices, sizes and counts. Historical local bytes and the pinned decoder remain authoritative for this store.

[Open primary source](#)

[3] **Databento issue board: GLBX.MDP3 historical channel-flush semantics** The linked board currently lists incorrect non-trade action/price/size on some flush records, with a trade-action distinction. An issue status does not certify repaired local files.

[Open primary source](#)

[4] **Databento: field conventions and GLBX.MDP3 supplement** Event completion, normalized schemas and unknown aggressor side. Also [field/flag definitions](#). Resolve historical applicability before use.

[Open primary source](#)

[5] **Databento: symbology** Volume-continuous selection uses previous-day volume rank; it is not a price-adjusted continuous instrument.

[Open primary source](#)

[6] **Gould & Bonart: queue imbalance and the next price move** Primary research motivating Q1. The paper's market and results do not establish an effect in GC.

[Open primary source](#)

[7] **Cont, Kukanov & Stoikov: price impact of order-book events** Primary research motivating best-level state pressure. Contemporaneous impact and out-of-sample prediction are separate claims.

[Open primary source](#)

[8] **CME Group: Gold futures specifications** GC contract size of 100 troy ounces and 0.10 price increment imply \$10 per tick. Applicable fees, contract mapping and live rules still require verification before execution.

[Open primary source](#)

External sources checked during preparation on 30 September 2026 (local date). No full book text was ingested. Proposed statistical thresholds, latency scenarios and workflow gates are design recommendations, not facts derived from these sources.